Topic 3, Lesson 3

Amazon Storage Services

Outline

- Overview of EBS, EFS, and S3 storage solutions
- Snowball and Snowmobile for data transfer
- Accessing S3 storage from EC2 instances

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Learning outcome

Storage Models: Explain object, block, and file storage paradigms and their fundamental trade-offs in cloud architectures.

Security & Durability: Apply AWS security best practices, durability patterns, and cost optimization strategies across storage services

Service Selection: Choose between S3, EBS, EFS, and FSx based on workload characteristics, performance requirements, and cost constraints

Overview of EBS, EFS, and S3 storage solutions

Real App story: FoodSnap

Running the App

FoodSnap runs on servers (EC2): Each server uses EBS like its own internal hard drive.

- Stores app, OS, and database (orders, users)

Scaling the App

As users increase, multiple servers are added.

- All servers use **EFS** like a **shared folder**
  - Store common files (menus, configs)

Storing User Data

Users upload food images and data:

- Stored in S3 like a cloud storage bucket
  - Unlimited, durable storage for images & backups

Accessing S3 from Servers

- EC2 servers fetch images from S3 when needed
- Also upload new images back to S3

⟶ Like app servers reading / writing data from cloud storage

Migrating Large Old Data

- FoodSnap has TBs / PBs of old data to move to AWS:
  - Snowball → Send physical device to transfer large data
  - Snowmobile → Use truck for extremely massive data transfer

The flow goes :

- EBS for server storage
- EFS for shared files S3 for user data
- EC2 ↔ S3 to access files
- Snowball/Snowmobile to move huge data to cloud”

Three Storage Models: Object, Block, File

Characteristic	Object Storage (S3)	Block Storage (EBS)	File Storage (EFS)
Data Organization	Flat namespace with metadata	Fixed-size blocks on volumes	Hierarchical directories
Access Protocol	HTTP/HTTPS REST API	SCSI, NVMe (OS-level)	NFS, SMB/CIFS
Typical Operations	PUT, GET, DELETE entire objects	Read/write blocks at byte offsets	Open, read, write, close files
Latency	Milliseconds to seconds	Sub-millisecond	Milliseconds
Throughput	Scales horizontally, massive aggregate	Limited by volume specs	Scales with parallel clients
Scaling	Virtually unlimited	Limited by volume size	Shared across clients
Use Cases	Media, backups, data lakes, web assets	Databases, boot volumes, transactional apps	Shared content, home directories, CMS

Amazon Elastic Block Store (EBS)

Amazon EBS is a persistent, high-performance block storage service for EC2.
It behaves like a virtual hard drive, designed for workloads needing consistent I/O and low latency

- Databases (MySQL, PostgreSQL, SQL Server, Oracle)
- Kubernetes persistent volumes (EKS/ECS)
- File systems and transactional workloads
- High-performance workloads requiring predictable IOPS

Types of EBS Volume and performance

- SSD, gp3, Provisioned IOPS SSD, HDD, Cold HDD, Throughput optimized HDD

Volume Type	Latency	IOPS Range	Throughput	Price Signal	Use Cases
gp3	Single-digit ms	3,000-16,000	125-1,000 MB/s	$	General purpose, boot volumes, dev/test
io2	Sub-millisecond	100-64,000	Up to 1,000 MB/s	$$$	Databases, latency-sensitive apps
io2 Block Express	Sub-millisecond	1,000-256,000	Up to 4,000 MB/s	$$$$	Largest databases, SAP HANA
st1	Low	500 (baseline)	40-500 MB/s	$	Big data, log processing (legacy)
sc1	Low	250 (baseline)	12-250 MB/s	$	Cold data, infrequent access (legacy)

Key Features of EBS

- Use gp3 as the default choice. It provides configurable IOPS and throughput independent of volume size, replacing older gp2 volumes. Use io2/io2 Block Express only for workloads that require sustained IOPS above 16,000 or sub-millisecond latency.

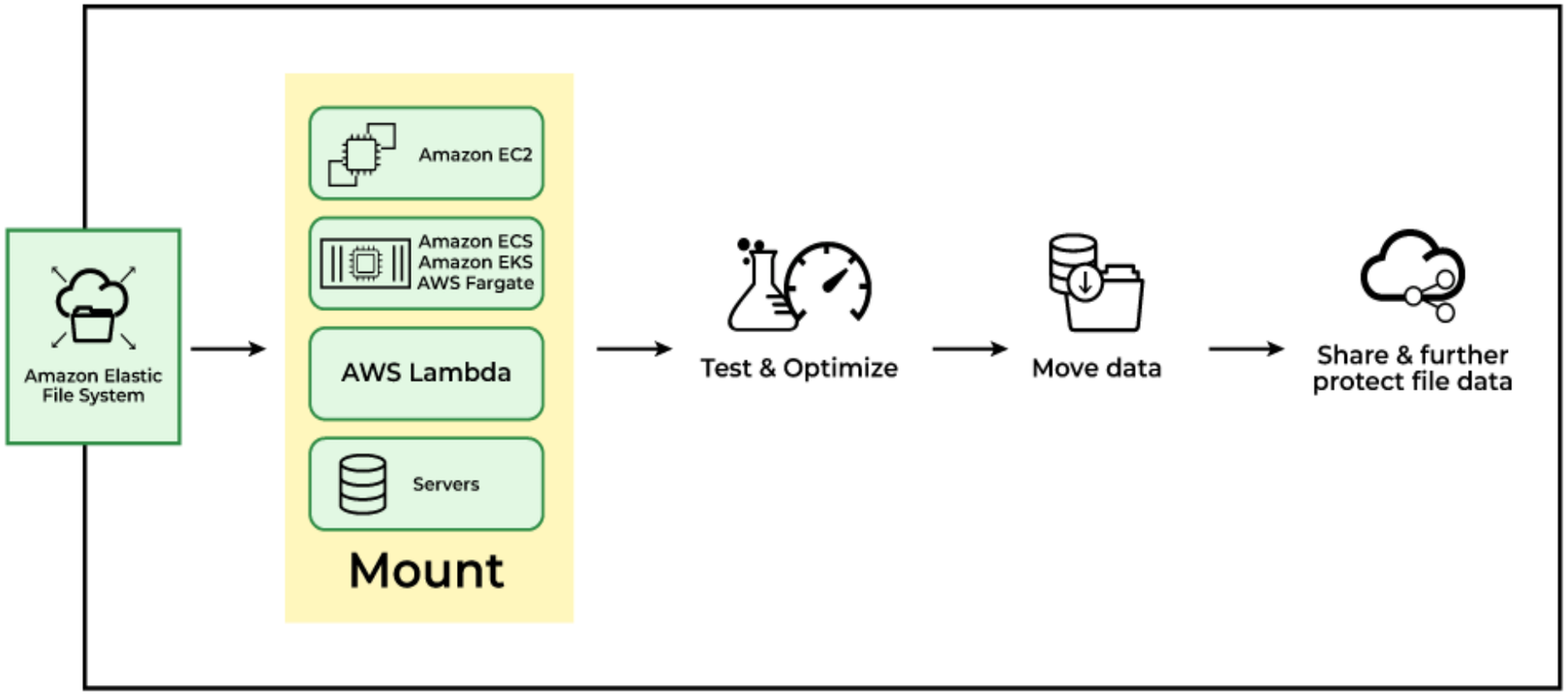
- Scalability & Elasticity: You can resize EBS volumes and adjust IOPS/throughput without downtime.
- Backup and Recovery (Snapshots): Snapshots act as backups of your volumes.
- Encryption: EBS encrypts data at rest and in transit via AWS KMS, with all snapshots and volumes staying encrypted.
- Durability: EBS volumes are highly available with automatic AZ replication, and io2 Block Express offers 99.999% durability.

Use case of EBS in database applications

- EBS can be used for databases as a root volume on EC2, storage for managed services like Amazon RDS/Aurora, or persistent storage for containerized databases.

Amazon Elastic file system (EFS)

- Amazon Elastic File System (EFS) is a serverless, fully managed, elastic file system that lets you share file data without provisioning or managing storage.
- It can be used with AWS Cloud services and on-premises resources, and is built to scale on demand to petabytes without disrupting applications.
- Unlike EBS (which is like a local hard drive for one server), EFS is a Network File System (NFS).
- It support NFSv4.1 and NFSv4.0) protocol.
- It means hundreds or thousands of EC2 instances, Lambda functions, and Containers can access the same files at the same time.

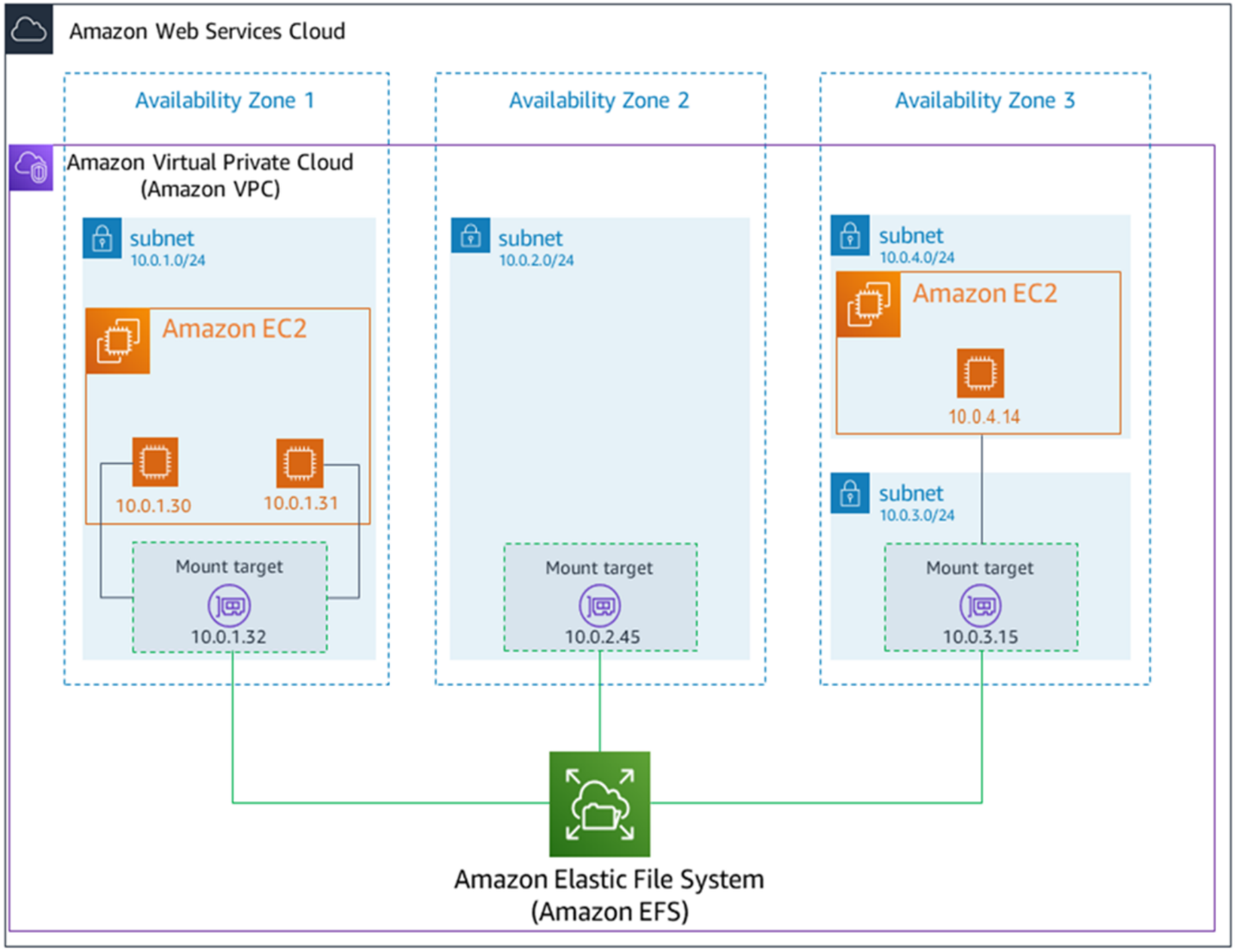


How Amazon EFS works with Amazon EC2

You can choose the EFS based on the I/Ops you are going to perform.

- **File System**: The top-level resource.
- Mount Targets: To access EFS, you create "Mount Targets" in your VPC subnets. These provide an IP address for your resources to connect to.
- **NFS Protocol**: Your EC2 instances use the standard NFSv4 protocol to mount the file system, making it look like a standard local directory (e.g., /mnt/efs).
- **Regional Availability**: Data written to EFS is automatically replicated across multiple Availability Zones (AZs) in a Region (unless you choose One Zone storage). This makes it highly durable.

- You can create and share Amazon EFS to EC2 instances easily. The details step are as:

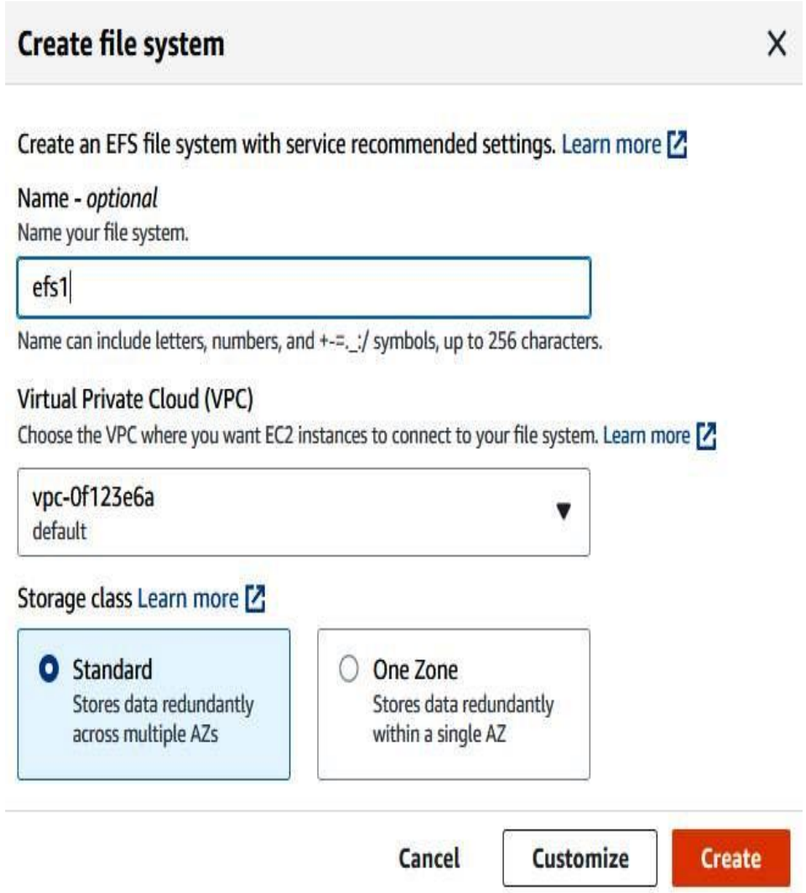


1. **Create an Amazon EFS filesystem:** Log into the AWS console, go to the EFS service, and create a filesystem.



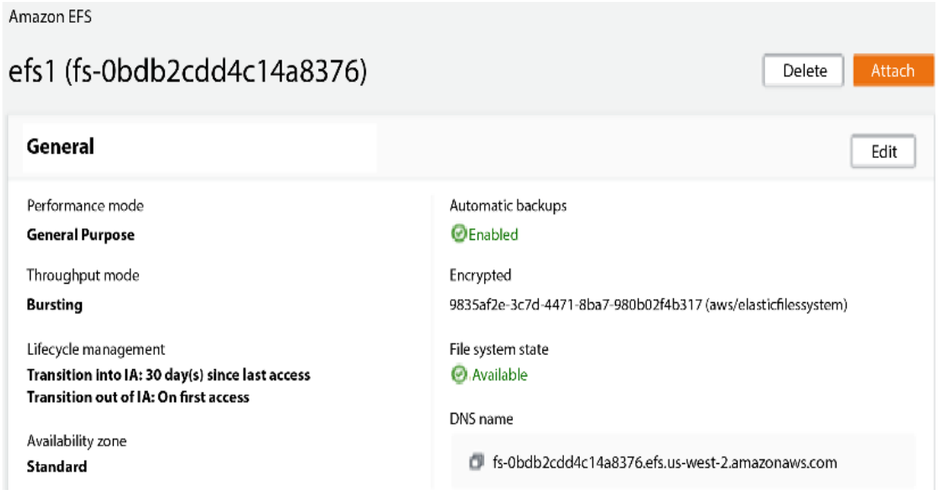
2. Click "Create file system": In the pop-up window, provide a name for the EFS filesystem and choose a storage class.

- A One Zone EFS filesystem stores data at the AZ level, and a standard filesystem stores data redundantly across multiple AZs in the same region.
- Note that there is no need to specify the size of the EFS filesystem since it scales elastically depending on the EC2 instances' demands on the shared filesystem.
- Click **Create**

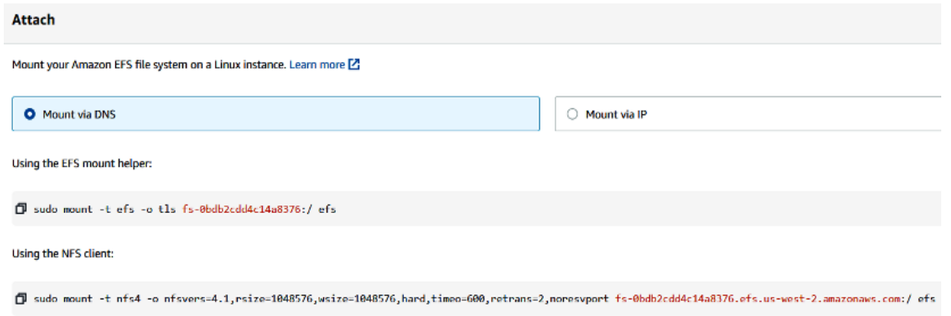


3. Attach the created EFS filesystem to a Linux instance:

- Review the EFS filesystem and click the **Attach**



4. Then click the **Using the NFS client tab**, then copy the Linux sudo mount command



5. Go to the Linux guest OS, use mkdir to create a new directory, and mount the EFS filesystem to the directory using the sudo mount command copied earlier. Then new filesystem is now ready for reading and writing.

```
[root@ip-172-31-26-122 /]# mkdir /efs1
[root@ip-172-31-26-122 /]# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,
retrans=2,noresvport fs-0bdb2cdd4c14a8376.efs.us-west-2.amazonaws.com:/ /efs1
[root@ip-172-31-26-122 /]# cd /efs1; cp /etc/hosts .
[root@ip-172-31-26-122 efs1]# ls
hosts
[root@ip-172-31-26-122 efs1]# touch x
[root@ip-172-31-26-122 efs1]# ls
hosts  x
```

6. Verify the EFS filesystem is shared from another Linux instance: Log into another Linux instance in the same region, use mkdir to create a directory, and **mount** *the same EFS filesystem* to the local directory. You will find that the filesystem is shared among the two EC2 instances with read and write.

- With a few clicks, we have created an EFS filesystem and shared it among two EC2 Linux instances in the same region.

```
[root@ip-172-31-31-184 ~]# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize
=1048576,hard,timeo=600,retrans=2,noresvport fs-0bdb2cdd4c14a8376.efs.us-west-2.
amazonaws.com:/ /efs2
[root@ip-172-31-31-184 ~]# ls /efs2
hosts  x
[root@ip-172-31-31-184 ~]#
```

Use Cases Of EFS

- **Secured file sharing:** You can share your files in every secured manner and in a faster and easier way and also ensures consistency across the system.
- **Web Hosting:** Well suited for web servers where multiple web servers can access the file system and can store the data EFS also scales whenever the data incoming is increased.
- **Modernize application development:** You can share the data from the AWS resources like ECS, EKS, and any serverless web applications in an efficient manner and without more management required.
- **Machine Learning and AI Workloads:** EFS is well suited for large data AI applications where multiple instances and containers will access the same data improving collaboration and reducing data duplication.

Performance & Throughput Modes

Performance Modes

- **General Purpose:** The default. Low latency. Best for web serving, CMS, and general file sharing.
- **Max I/O:** Higher latency, but massive throughput. Best for Big Data analytics and media processing with hundreds of parallel threads.

Throughput Modes

- **Elastic (Recommended):** The modern default. Throughput scales automatically with your workload. You pay for what you use. No guessing required.
- **Bursting:** Throughput is tied to your storage size. You earn "burst credits" when idle and spend them when active. Good for small, spiky workloads.
- **Provisioned:** You pay for a fixed amount of throughput (MB/s) regardless of your storage size. Useful if you have a tiny dataset that needs massive speed.

Security Best Practices

Security Groups: EFS has its own Security Group. You must allow Inbound NFS (Port 2049) from the Security Group of your EC2 instances/Lambda.
IAM Authorization: Don't rely just on network security. Use IAM policies to control which clients can mount the file system (e.g., prevent Root access).
Encryption: Always enable encryption At Rest (managed by KMS) and In Transit (using TLS when mounting).

Storage Classes & Lifecycle Management

- EFS offers four storage classes to optimize costs.
- You typically don't select these manually;
- you use **Lifecycle Management** to move files automatically.

- **EFS Standard**
- **EFS Standard-IA (Infrequent Access)**
- **EFS One Zone**
- **EFS One Zone-IA**

AWS Simple Storage Service (S3)

- Amazon Simple Storage Service (S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance.
- It is the backbone of the AWS ecosystem, used by everyone from tiny startups to massive enterprises like Netflix and Airbnb to store petabytes of data.
- Unlike file storage (like EFS) or block storage (like EBS), S3 stores data as Objects within Buckets.
- Virtually unlimited storage
  - Single object is limited to 5 TB
- Designed for 11 9s of durability
- Granular access to bucket and objects



Core Architecture: Buckets and Objects

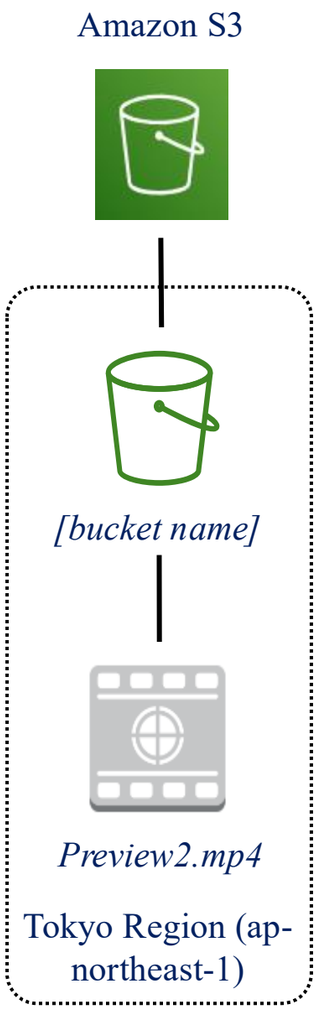
1. Bucket: A container for objects.
- **Global Namespace:** Bucket names must be unique across all AWS accounts worldwide (like a DNS name).
- **Region Specific:** You create a bucket in a specific region (e.g., us-east-1), and your data never leaves that region unless you explicitly move it.

2. Object: The fundamental entity stored in S3.
- **Key:** The name of the object (e.g., photos/vacation.jpg).
- **Value:** The data itself (bytes).
- **Metadata:** Name-value pairs describing the object (e.g., Content-Type: image/jpeg).
- **Size:** Objects can be from 0 bytes up to 5 Terabytes.

S3 Storage Classes

Storage Class	Access Frequency	Availability Zones	Use Case
S3 Standard	Frequent	>= 3	General-purpose storage, static websites, cloud apps.
S3 Intelligent-Tiering	Unknown/Changing	>= 3	Data with unpredictable access patterns. Automatically moves data between tiers to save costs.
S3 Standard-IA	Infrequent	>= 3	Long-lived data accessed less than once a month (e.g., backups, DR).
S3 One Zone-IA	Infrequent	**1**	Non-critical, reproducible data (e.g., secondary backups). Cheaper but less durable.
S3 Glacier Instant	Rare (Quarterly)	>= 3	Archives that need millisecond access (e.g., medical records).
S3 Glacier Flexible	Rare (Yearly)	>= 3	Archives where retrieval time of minutes/hours is acceptable.
S3 Glacier Deep Archive	Very Rare	>= 3	Long-term retention (7-10 years) for compliance. Lowest cost.

Amazon S3 bucket URLs (two styles)



To upload your data:

1. Create a **bucket** in an AWS Region.
2. Upload almost any number of **objects** to the bucket.

Bucket path-style URL endpoint:

https://s3.ap-northeast-1.amazonaws.com/bucket-name

Region code **Bucket name**

Bucket virtual hosted-style URL endpoint:

https:// bucket-name.s3-ap-northeast-1.amazonaws.com

Bucket name Region code

Amazon S3 Usage

1. **Data Storage**
2. **Backup and Recovery**
3. **Hosting Static Websites**.
4. **Data Archiving:** Amazon S3 Glacier service integration helps as a cost-effective solution for long-term data storing.
5. **Big Data Analytics:** Amazon S3 is often considered as data lake because of its capacity to store large amounts of both structured and unstructured data offering seamless integration with other AWS Analytics and AWS Machine Learning Services.

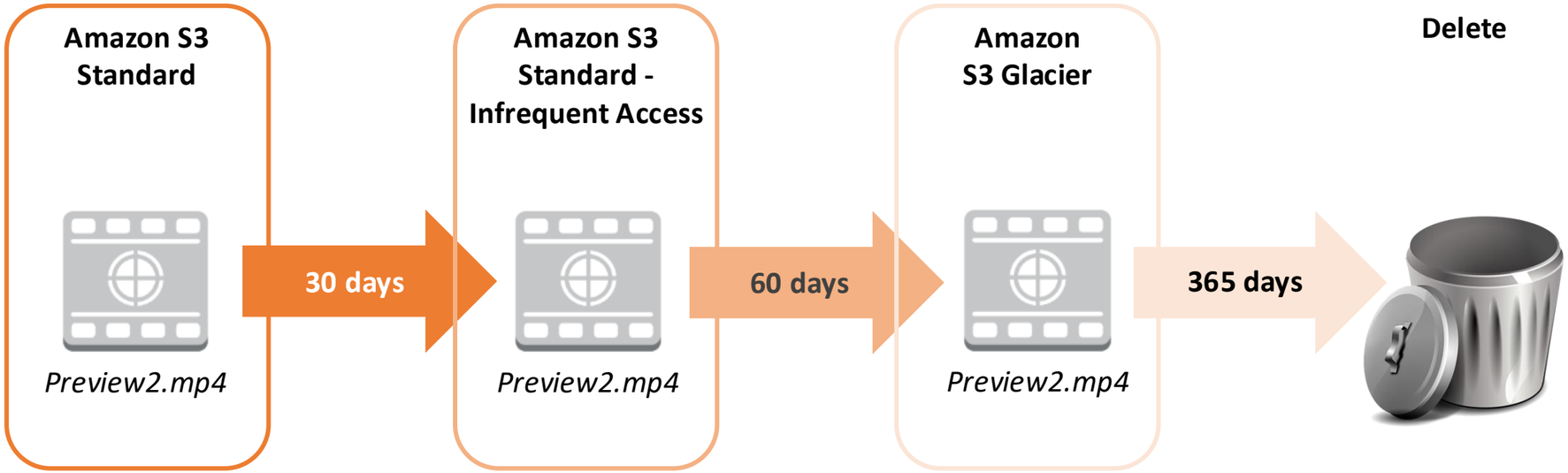
Amazon S3 pricing

- Pay only for what you use, including –
  - GBs per month
  - Transfer OUT to other Regions
  - PUT, COPY, POST, LIST, and GET requests
- You do not pay for –
  - Transfers IN to Amazon S3
  - Transfers OUT from Amazon S3 to Amazon CloudFront or Amazon EC2 in the same Region

S3 Security Capabilities

- Data Consistency
- S3 Versioning
- S3 Lifecycle Management
- Security & Access Control
- Encryption

Amazon S3 lifecycle policies enable you to delete or move objects based on age.



Security with Amazon S3 Glacier



Amazon S3 Glacier



Control access with IAM



Amazon S3 Glacier encrypts your data with AES-256



Amazon S3 Glacier manages your keys for you

Use an Amazon S3 Bucket

1. Login into the Amazon account with your credentials and search form S3 and click on the S3.

2. Now click on the option which is **"Create bucket"** and configure all the options which are shown while configuring.

3. After configuring the AWS bucket now upload the objects into the buckets based upon your requirement.

- using the AWS console or by using AWS CLI following is the command to upload the object into the AWS S3 bucket.
- *aws s3 cp <local-file-path> s3://<bucket-name>/*

4. You can control the permissions of the objects which was uploaded into the S3 buckets and also who can access the bucket.

- You can make the bucket public or private by default the S3 buckets will be in private mode.

5. You can manage the S3 bucket lifecycle management by transitioning.

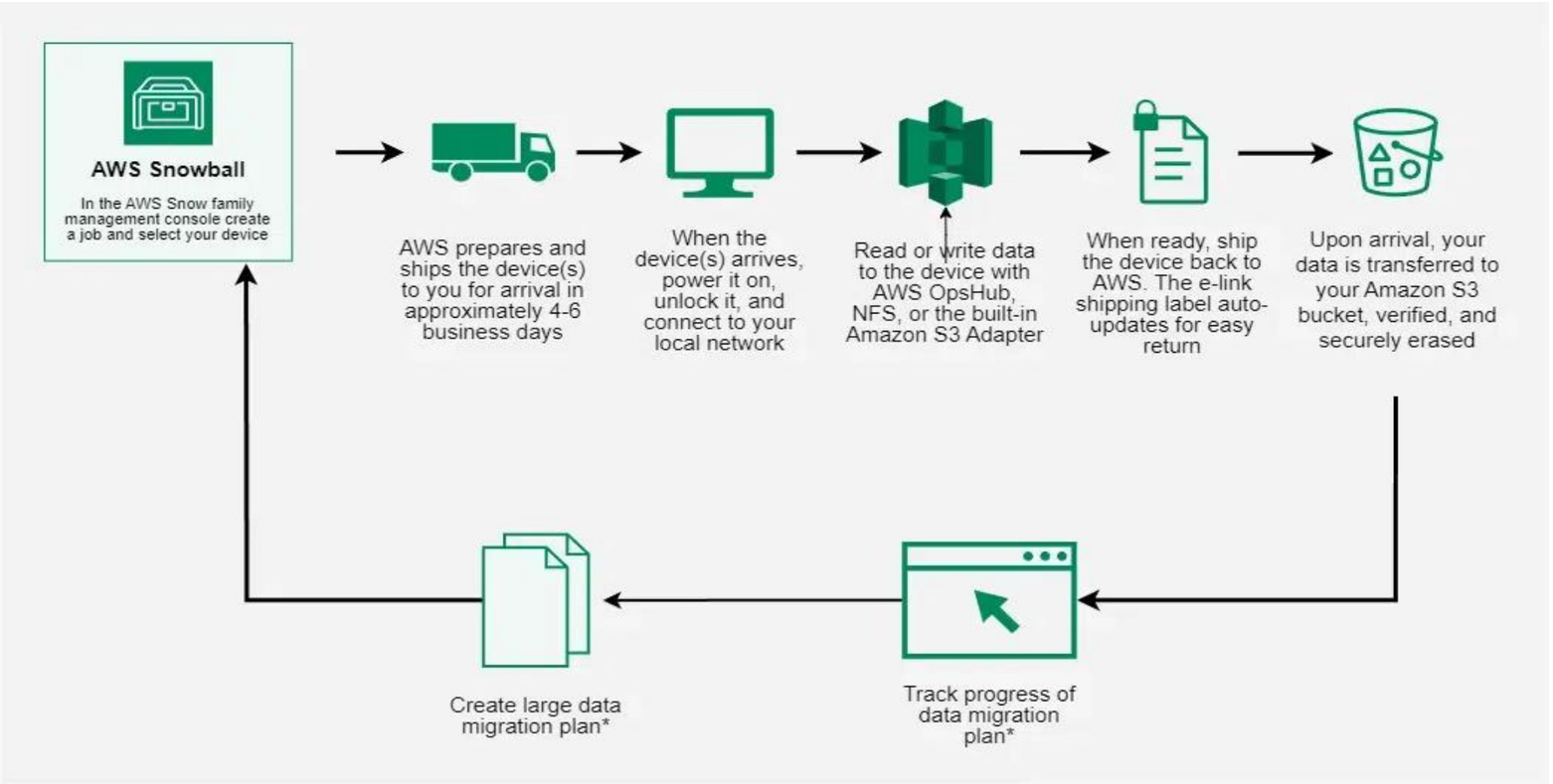
- Based upon the rules that you defined S3 bucket will be transitioning into different storage classes based on the age of the object which is uploaded into the S3 bucket.

6. You need to turn to enable the services to monitor and analyze S3.

- You need to enable the S3 access logging to record who was requesting the objects which are in the S3 buckets.

What is Snowball

- AWS Snowball is a service from Amazon Web Services that provides a secure and efficient way to transfer large amounts of data to and from the cloud.
- It uses a rugged physical device called a Snowball to move data between on-premises environments and AWS.
- It helps businesses handle massive data transfers and take advantage of cloud storage and processing.



How AWS snowball works

- Order an AWS Snowball device from AWS.
- AWS ships the physical device to your location.
- Connect the device to your local network.
- Transfer your data onto the Snowball.
- Ship the device back to AWS.
- AWS uploads the data to your cloud storage.

Use cases

- Migrate data at petabyte-scale: Move large data to the cloud when network is limited.
- Process and analyze data locally: Run AMIs and AWS Lambda on Snowball Edge for ML and other apps.
- Optimize manufacturing data: Collect and analyze on-site factory data to refine processes and improve safety, efficiency, and productivity.

What is AWS Snowmobile

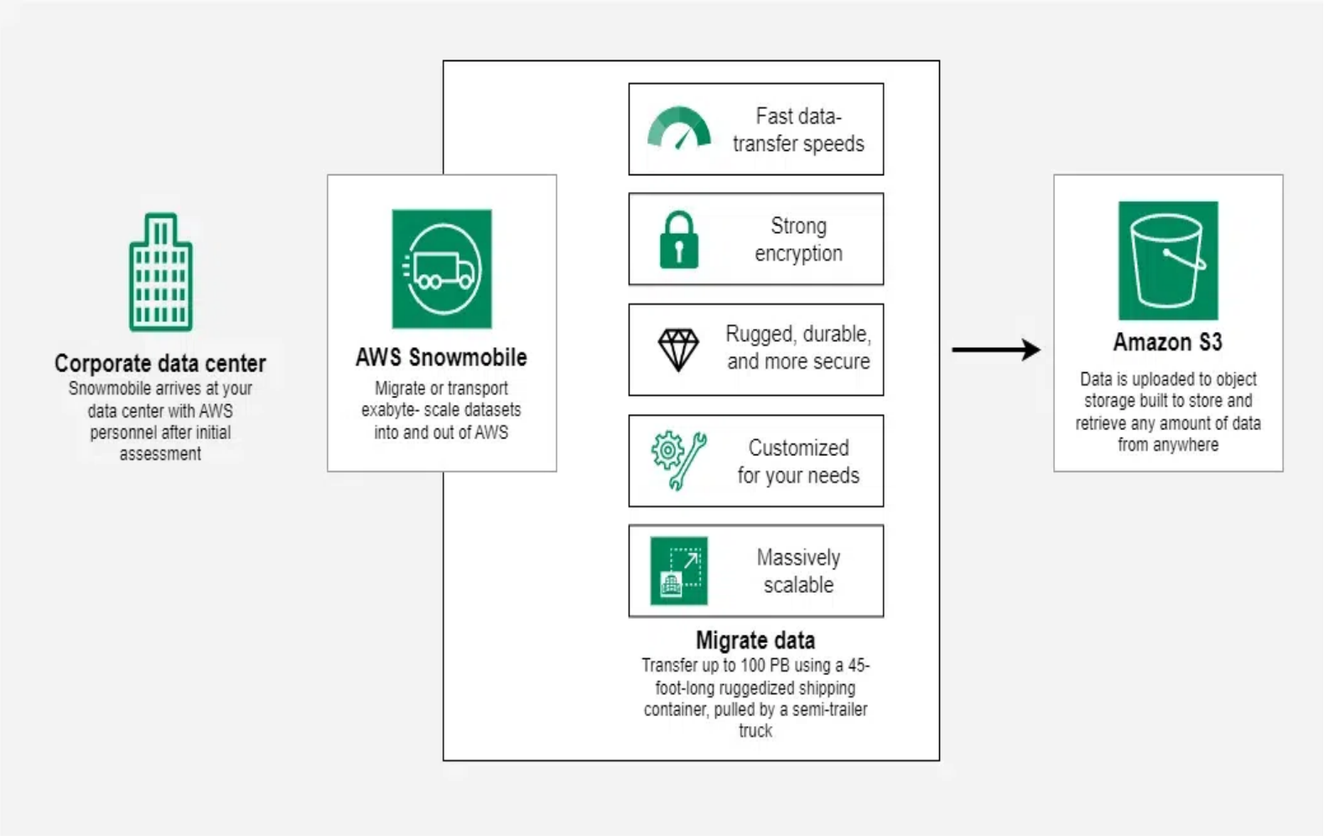
- AWS Snowmobile is a physical data transfer service designed for large-scale data migrations to the AWS cloud.
- It is essentially to secure 45-foot long shipping container that can transfer up to **100 petabytes** of data in a single shipment.
- AWS Snowmobile is ideal for industries needing to move vast amounts of data such as media, entertainment or government sectors.
- It offers high-level security Including encryption GPS tracking and security personnel to ensure safe transport of your data to AWS data centers.

Use cases

- Data center migration
- Disaster recovery
- Video libraries
- Research data transfer
- Government and security agencies

How does AWS Snowmobile Work

- AWS Snowmobile works by physically transporting large amounts of data from your location to AWS data centers.



Accessing S3 storage from EC2 instances

- To have an EC2 instance access S3 buckets, we need to assign an IAM role to the EC2 instance.
- An IAM role can be assigned with permissions to access other AWS resources, such as reading an Amazon Simple Storage Service (S3) bucket. By attaching the IAM role to an EC2 instance, all the applications running on the EC2 instance will have the permissions of that role.

1. Create an IAM Role for EC2

I. Open **AWS Identity and Access Management** in the console
II. Go to **Roles → Create role**
III. Choose:
    I. Trusted entity: **AWS service**
    II. Use case: **EC2**
IV. Click **Next**
V. Attach permissions
    I. For quick setup: choose AmazonS3ReadOnlyAccess
    II. Or create a custom policy (recommended for tighter control)
VI. Name the role (e.g., EC2-S3-Access-Role)
VII. Click **Create role**

2. **Attach Role to EC2 Instance**

- Open EC2 Console
- Go to Instances
- Select your instance
- Click Actions → Security → Modify IAM role
- Choose the role you created
- Click Update IAM role

3. Verify Access via EC2 (Browser or Session)

- Open S3 Console
- Navigate to your bucket
- If your IAM role and permissions are correct, your applications on EC2 will also be able to access it.

4. **(Optional) Restrict to Specific Bucket**

In IAM → edit the role → add inline policy like:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
     "Action": ["s3:GetObject", "s3:ListBucket"],
     "Resource": ["arn:aws:s3:::your-bucket",
                  "arn:aws:s3:::your-bucket/*”
                  ]
     }
  ]
}
```

5. (Important) Networking Check

- If your EC2 instance is in a private subnet:
  - Create a **VPC Endpoint for S3** in the VPC console
- Otherwise:
  - Ensure it has internet access (IGW or NAT)

Further reading and resources

- Chapter 2:Amazon Cloud Storage Services. The Self-Taught Cloud Computing Engineer by Dr. Logan Song
- Amazon Elastic File System Documentation. https://docs.aws.amazon.com/efs/latest/ug/how-it-works.html
- AWS Elastic File System(EFS) https://www.geeksforgeeks.org/cloud-computing/introduction-to-aws-elastic-file-systemefs/
- AWS Simple Storage Service (AWS S3) https://www.geeksforgeeks.org/devops/introduction-to-aws-simple-storage-service-aws-s3/
- What is Amazon S3? https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html
- Snowball. https://aws.amazon.com/snowball/
- Snow Family in AWS https://www.geeksforgeeks.org/devops/aws-snow-family/
- Use Amazon S3 with Amazon EC2 instances. https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AmazonS3.html